Ground State Degeneracy in the Levin-Wen Model

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Topological Quantum Matter

- Time Reversal broken:

  Integer Quantum Hall Effect

  Fractional Quantum Hall Systems

  (Topological quantum number: Chern number)

- Time Reversal invariant:

  Topological Insulators

  Quantum Spin Liquids

  Topological Superconductors

  Topological Superfluids ……..

  (Topological quantum number: Z_2 number)

Landau's Theory of Classical Phase Transitions

- **Classical Phase Transitions:**

  - ✓ **Driven by thermal fluctuations**
  - ✓ **Associated with symmetry breaking**
  - ✓ **Characterized by (local) order parameter(s)**

Examples: crystals

ferromagnetism

superfluids

superconductivity etc

Topological Phases: beyond the Landau's Paradigm

- Novel phases at T=0: due to quantum effects
- No symmetry breaking and order parameter
- Many-body state with a topological number,

  that is robust against weak disorders and interactions

- Topology dependent ground state degeneracy
- Emergent exotic quantum numbers for excitations:

  fractional charge, fractional statistics,

  fractional particle number etc
- Bulk/Edge(Surface) Correspondence (Holography)

Topological Matter (Order, Phases)

- What is a Topological State of Matter?

  (Answer of physicists)

  It has following properties:

  - Gapped Energy Spectrum (necessary condition)
  - The (many-body) state has a Topological Quantum Number:

    robust against (weak) disorders and interactions
  - Characteristic Property: No local degree of freedom
    - Ground states look the same at all length scales.
    - All ground states look the same locally: Local measurements cannot distinguish between any two ground states.
  - Robust Ground State Degeneracy
  - Fractionalization of Particle-like Excitations: charge and statistics,
  - Gapless edge (surface) excitations: Bulk-Edge Correspondence

Topological Matter (Order, Phases)

- What is a Topological Phase?
  (Answer of mathematicians)

  It is a state of matter that can be described by a
  topological quantum field theory (TQFT)

  - For a spatially two-dimensional system,
    TQFT is three-dimensional (space-time)

  - Atiyah's Axioms for TQFT are formulated for manifolds

Discrete Models for Topological Matter

Physicists's Preferences:

- How to formulate a model on a lattice or a graph?

  Easier to solve exactly or to do numerical simulation
- How to formulate in terms of a Hamiltonian?

  Closer to what has been familiar to physicists

- Kitaev's Toric Code Model:

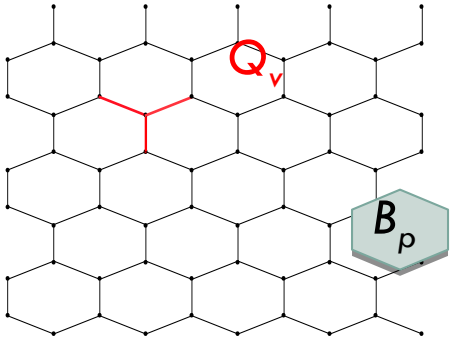
  Has excitations with fractional (braid) statistics
- Levin-Wen's Model:
  - String-net condensed states
  - "Fixed point" wave functions that look the same at all length scales.

Kitaev's Toric Code Model (Z_2 gauge theory)

- Hamiltonian

$$\hat{H} = -\sum_v \hat{Q}_v - \sum_p \hat{B}_p$$

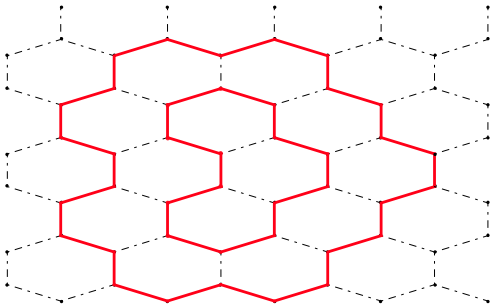
$$\hat{Q}_v = \prod_{j \in \text{star}(v)} \sigma_j^z \quad \hat{B}_p = \prod_{j \text{ around } p} \sigma_j^x$$



- Ground state: superposition of

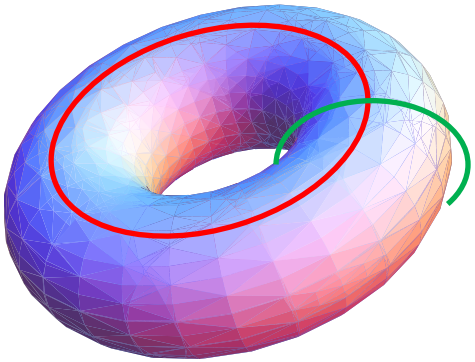
  all configurations of closed strings

string-net condensation: $|\Phi\rangle = \sum_{\text{loops } X} |X\rangle$



$$\begin{cases} |\sigma_z = +1\rangle & : \text{occupied string} \\ |\sigma_z = -1\rangle & : \text{empty string} \end{cases}$$

- GSD = 4 on a torus



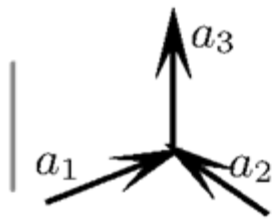
A. Yu. Kitaev, Annals Phys. 303 (2003) 2-30

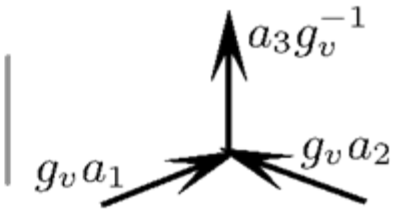
X-G. Wen, *Quantum Field Theory of Many Body Systems*, Oxford Univ. Press, Oxford, 2004.

Generalized Kitaev's model (with any finite group G)

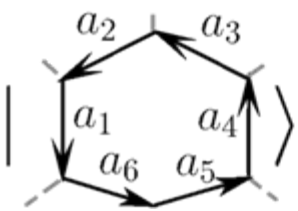
$$\hat{H} = -\sum_v \hat{Q}_v - \sum_p \hat{B}_p$$

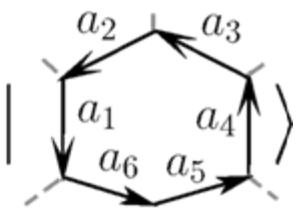
- Hilbert space: tensor product of group algebras C[G] on edges
- Q term prefers gauge invariance:

$\hat{Q}_v$ 

$= |G|^{-1} \sum_{g_v \in G}$ 

- *B* term prefers flat connections:

$\hat{B}_p$ 

$= \delta_{a_1 a_2 a_3 a_4 a_5 a_6, e}$ 

- E.g., in the Toric Code model,

$Z_2 = \{e, \sigma_x\}$

$\mathbb{C}[Z_2]$ spanned by $\{|e\rangle, |\sigma_x\rangle\}$

where $|e\rangle = |\sigma_z = +1\rangle, |\sigma_x\rangle = |\sigma_z = -1\rangle$

Ground state degeneracy in Kitaev's model

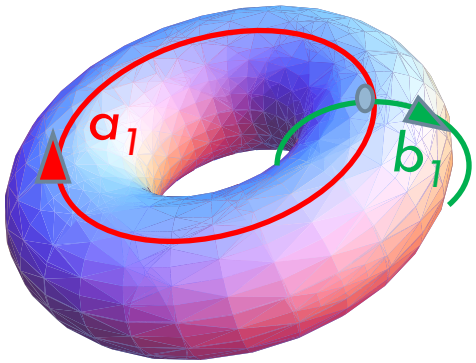
- Counting configurations of flat connections (modulo gauge transformations):

$$\mathrm{GSD} = \left|\frac{\mathrm{Hom}(\pi_1(\mathcal{M}), G)}{G}\right|$$

(with G in the denominator: diagonal conjugacy classes)

- Ground states labeled by non-contractible holonomies

E.g., holonomies a_1 b_1 along two non-contractible loops on a *torus*:



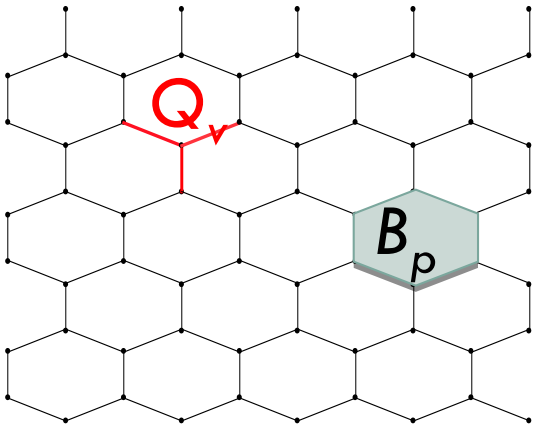
Ground-state subspace

$$= \mathrm{span}\{|a_1, b_1\rangle \,|a_1, b_1 \in G, a_1 b_1 a_1^{-1} b_1^{-1} = e\}$$

modulo the equivalence

$$|a_1, b_1\rangle \sim \left|g a_1 g^{-1}, g b_1 g^{-1}\right\rangle \text{ for any } g \in G$$

Levin-Wen model (1): on a trivalent graph



- Hamiltonian (in General)

$$\hat{H} = -\sum_v \hat{Q}_v - \sum_p \hat{B}_p$$

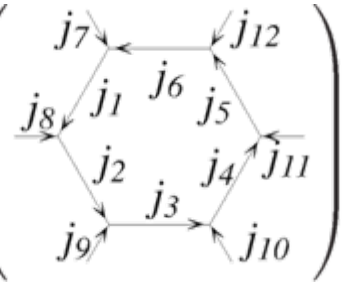
- Hilbert space:

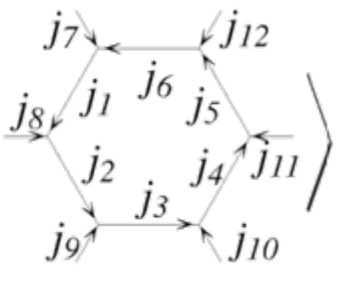
  {configurations of strings on edges}

- Ground states: string-net condensed states

$$\begin{cases} \hat{Q}_v |\Phi_a\rangle = |\Phi_a\rangle \\ \hat{B}_p |\Phi_a\rangle = |\Phi_a\rangle \end{cases}$$

$$|\Phi_a\rangle = \sum_{j_1, j_2, \ldots} \Phi_a \left(\right. \quad \left. \right| \quad \left. \right\rangle$$

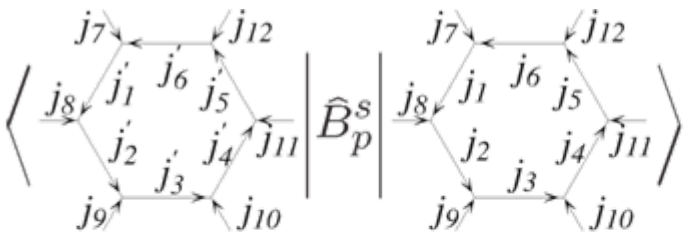




M. A. Levin, X-G. Wen, Phys.Rev. B71 (2005) 045110

The Levin-Wen model (2): 6*j* symbols as input data

- Q terms imposes the branching rules.
- Each B term is a sum: $\hat{B}_p = D^{-1} \sum_s d_s \hat{B}_p^s$



$$= v_{j_1} v_{j_2} v_{j_3} v_{j_4} v_{j_5} v_{j_6} v_{j_1'} v_{j_2'} v_{j_3'} v_{j_4'} v_{j_5'} v_{j_6'}$$

$$G^{j_7 j_1^* j_6}_{s^* j_6' j_1'^*} G^{j_8 j_2^* j_1}_{s^* j_1' j_2'^*} G^{j_9 j_3^* j_2}_{s^* j_2' j_3'^*} G^{j_{10} j_4^* j_3}_{s^* j_3' j_4'^*} G^{j_{11} j_5^* j_4}_{s^* j_4' j_5'^*} G^{j_{12} j_6^* j_5}_{s^* j_5' j_6'^*}$$

where

$$D = \sum_j d_j^2, \; v_j = \sqrt{d_j}$$

G is the symmetrized 6j symbol. It may be recognized as the Racah 6j symbol in group representation theory.

symmetry: $G^{ijm}_{kln} = G^{mij}_{nk^*l^*} = G^{klm^*}_{ijn^*} = (G^{j^*i^*m^*}_{l^*k^*n})^*$

pentagon id: $\sum_n d_n G^{mlq}_{kp^*n} G^{jip}_{mns^*} G^{js^*n}_{lkr^*} = G^{jip}_{q^*kr^*} G^{riq^*}_{mls^*}$

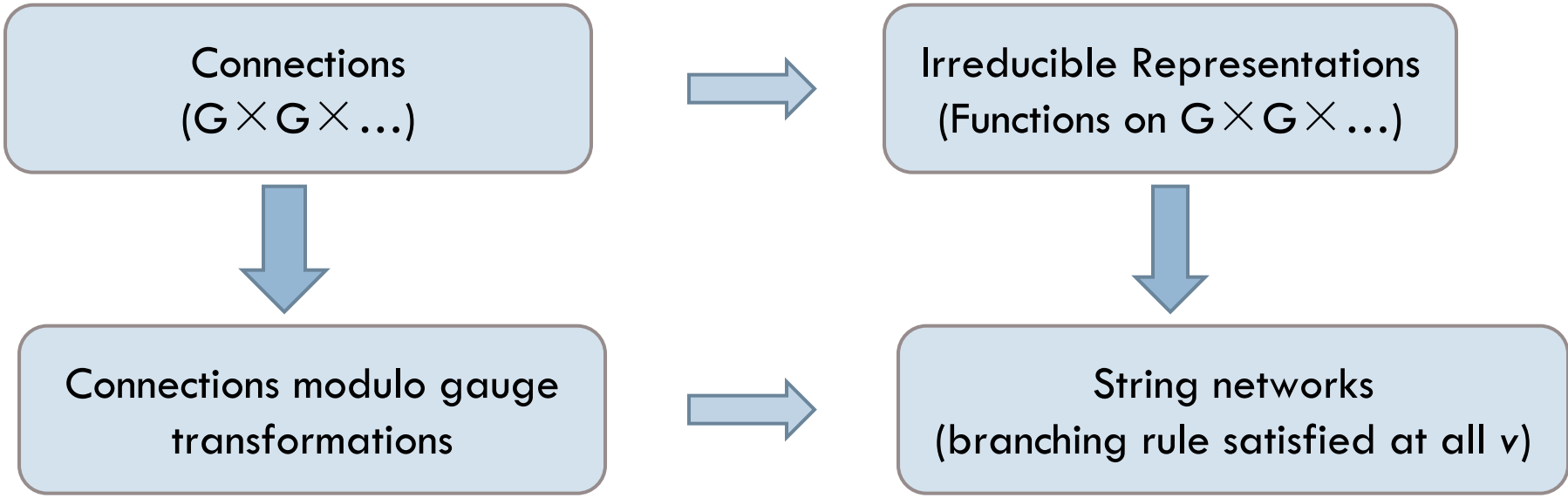
orthogonality: $\sum_n d_n G^{mlq}_{kp^*n} G^{l^*m^*i^*}_{pk^*n} = \frac{\delta_{iq}}{d_i} \delta_{mlq} \delta_{k^*ip}$

Map the Kitaev's model to the LW model: Peter-Weyl theorem adapted on a graph

- Group G ➔ Functions on G:

$$\{|a\rangle\}_{a\in G} \mapsto \{|\rho_{\alpha\beta}\rangle := \sqrt{\frac{\dim(\rho)}{|G|}} \sum_{a\in G} \rho^*_{\alpha\beta}(a)|a\rangle\}_{\rho\in \mathrm{Irreps}}$$

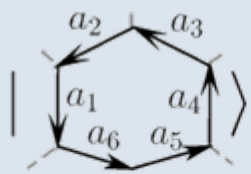
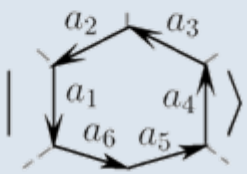
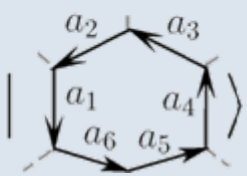
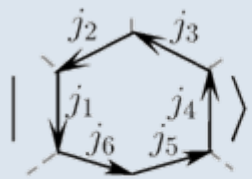
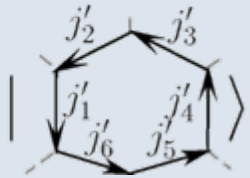
- Conjugacy classes ➔ Characters (Wilson loops)
- On a graph (embedded into a Riemannian surface)



Span Hilbert space
In Kitaev's model

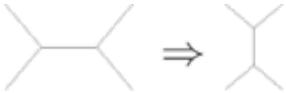
Span Hilbert space
in LW model

Map Kitaev's model to the LW model: Peter-Weyl theorem adapted on a graph

	Kitaev's model	LW model
Hilbert space	Groups on edges {a1, a2, …}	Functions on groups on edges {f(a1, a2, …)}
constraint $Q_v=1$ for all v	{a1, a2, …} Modulo gauge transformations	Gauge invariant functions = string networks (with branching rules)
Under the constraint $Q_v=1$	$\hat{B}_p$  $= \delta_{a_1a_2a_3a_4a_5a_6,e}$  $= \frac{1}{\lvert G\rvert}\sum_{s\in Rep(G)} d_s tr_s(a_1a_2a_3a_4a_5a_6)$ 	$\hat{B}_p$  $= \frac{1}{\lvert G\rvert}\sum_{s\in Rep(G)} d_s \sum_{\{j'\}} (B_p^s)^{j_1',j_2',j_3',j_4',j_5',j_6'}_{j_1,j_2,j_3,j_4,j_5,j_6}$ 

In terms of Racah 6*j* symbols!

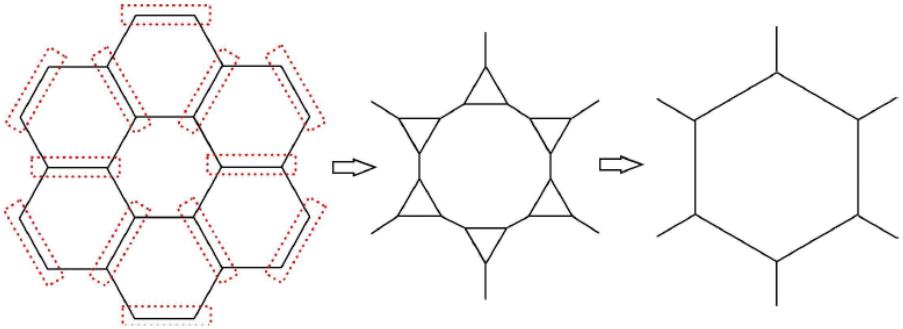
Mutations of trivalent graphs

f_1.  , for any edge;

f_2. , for any vertex.

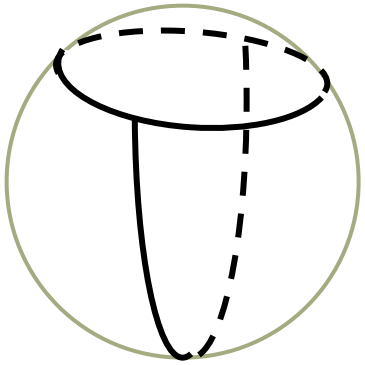
f_3. , for any triangle structure.

Example:
A mutation of honeycomb lattice

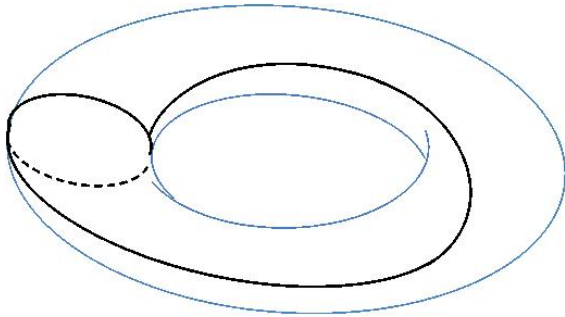


Basic rules of mutations (Pachner moves)

By repeating the basic rules of mutations, all trivalent graphs can be mutated to the simplest ones.



sphere



torus

Mutation transformations

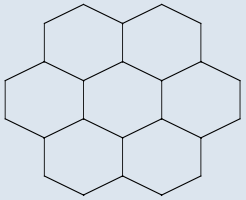
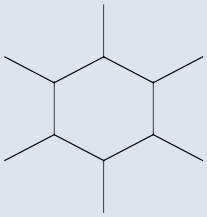
Main idea:

Ground states are kept "the same" during a mutation of the graphs, if

1). we can find a 1-to-1 transformation induced by the mutation;

2). the induced transformation does not depend which way we mutate the graph. It is unique!

Mutation between any two graphs

	Graph Γ	$\Gamma \rightarrow \Gamma'$	Graph Γ'
Graphs before and after the mutation		Deformation of graphs	
Hilbert space of ground states	$\mathcal{H} = \{\Phi_a\}$	1 to 1 map	$\mathcal{H}' = \{\Phi'_a\}$

There is an 1-to-1 map between ground states, induced by any mutation of the graphs: $\hat{T} : \mathcal{H} \rightarrow \mathcal{H}'$

Mutation Transformations: "path"-independence

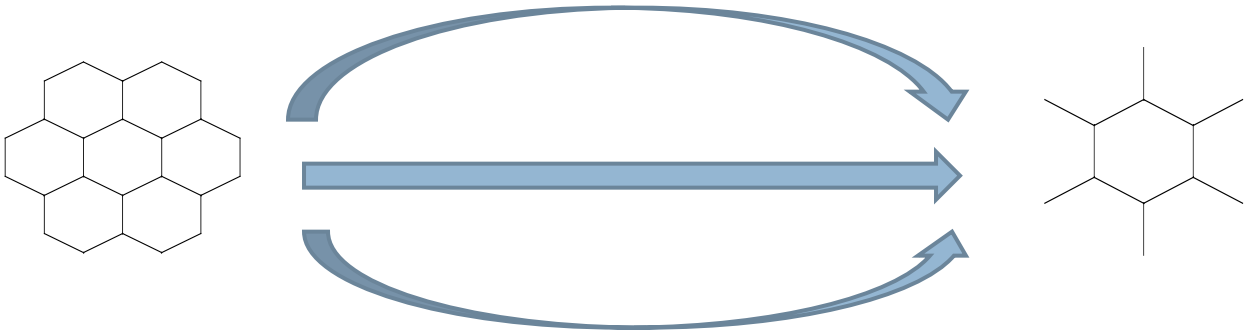
Main idea:

Ground states are kept "the same" during a mutation of the graph, if

1). we can find a 1-to-1 transformation induced by the mutation;

2). the induced transformation does not depend which way we mutate the graph. It is unique!

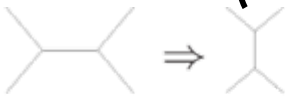
- There are many ways to mutate one graph to another graph



- Different ways of mutations induce the same transformation!

Mutation transformations: Definitions and Realizations

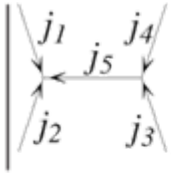
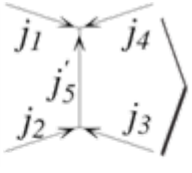
- Basic mutations (Pachner, 1978)

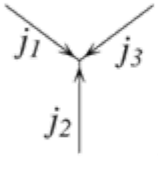
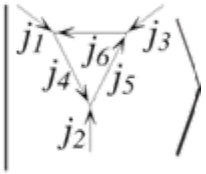
f_1.  , for any edge;

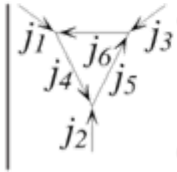
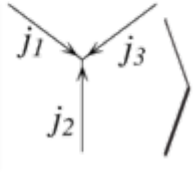
f_2. , for any vertex.

f_3. , for any triangle structure.

- Unitary transformations (our own work)

$\hat{T}_1$:  $\rightarrow \sum_{j_5'} v_{j_5} v_{j_5'} G^{j_1 j_2 j_5}_{j_3 j_4 j_5'}$ 

$\hat{T}_2$:  $\rightarrow \sum_{j_4 j_5 j_6} \frac{v_{j_4} v_{j_5} v_{j_6}}{\sqrt{D}} G^{j_2 j_3 j_1}_{j_6^* j_4 j_5^*}$ 

$\hat{T}_3$:  $\rightarrow \frac{v_{j_4} v_{j_5} v_{j_6}}{\sqrt{D}} G^{j_3^* j_2^* j_1^*}_{j_4^* j_6 j_5^*}$ 

Understand the ground states: How to define a topological phase

Unitary mutation transformations

"path"-independence

A discrete TQFT

(the induced transformations is the functor in Atyah's definition of TQFT)

triangulation-independence of the path integral

Fixed points of RG flow

(Not only under the scaling transformations, but under all mutations that preserve the topology of the surface)

Fixed-points of the *ground states*, not the Hamiltonian

GSD as a topological invariant

- Understand topological order → Calculate GSD
- LW model as a Hamiltonian version of a discrete TQFT:

GSD Of LW model ⟷ partition function Of (2+1)-dimTQFT ⟷ Topological invariant of 3d manifold

Ground state degeneracy

Algebraic way	Topological properties
Constructed by local tensors (6j symbols)	Depends only on the topology of surface

Z. Kadar, A. Marzuoli, M. Rasetti, Advances in Mathematical Physics, Volume 2010 (2010)

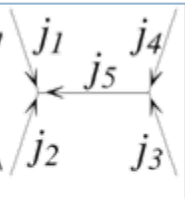
Find GSD from local G symbols

$$\hat{Q}_v \hat{Q}_v = \hat{Q}_v \ , \ \hat{B}_p \hat{B}_p = \hat{B}_p$$

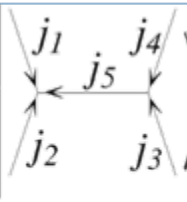
$\prod_v \hat{Q}_v \prod_p \hat{B}_p$ is the projector onto the ground states

$\mathrm{GSD} = \mathrm{tr}(\prod_v \hat{Q}_v \prod_p \hat{B}_p)$ In terms of 6*j* symbols

$$\mathrm{GSD} = \sum_{j_1 j_2 j_3 j_4 j_5 j_6 \cdots} \Big\langle$$



$$\Big| \left(\prod_v \hat{Q}_v \prod_p \hat{B}_p\right) \Big|$$



$$\Big\rangle$$

$$= D^{-P} \sum_{s_1 s_2 s_3 s_4 \cdots} d_{s_1} d_{s_2} d_{s_3} d_{s_4} \cdots$$

$$\sum_{j'_1 j'_2 j'_3 j'_4 j'_5 \cdots} d_{j'_1} d_{j'_2} d_{j'_3} d_{j'_4} d_{j'_5} \cdots \sum_{j_1 j_2 j_3 j_4 j_5 \cdots} d_{j_1} d_{j_2} d_{j_3} d_{j_4} d_{j_5} \cdots$$

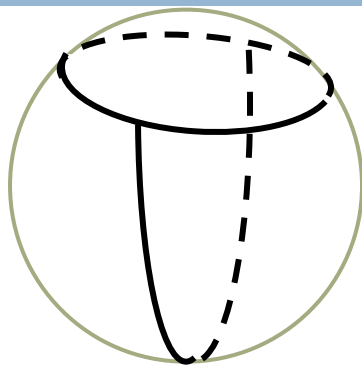
$$\left(G^{j_2 j_5 j_1}_{s_1^* j'_1 j'_5} G^{j'_1 j_2 j'_5}_{s_2^* j_5 j'_2} G^{j_5 j'_1 j'_2}_{s_3^* j_2 j_1} \right) \left(G^{j_3 j_4 j_5^*}_{s_1^* j_5'^* j'_4} G^{j'_4 j_5'^* j_3}_{s_2^* j'_3 j_5^*} G^{j_5^* j'_3 j'_4}_{s_4^* j_4 j_3} \right) \cdots$$

P: total number of vertices

Examples: on a sphere/Torus

- Sphere
  - GSD = 1 (With any detailed structure)

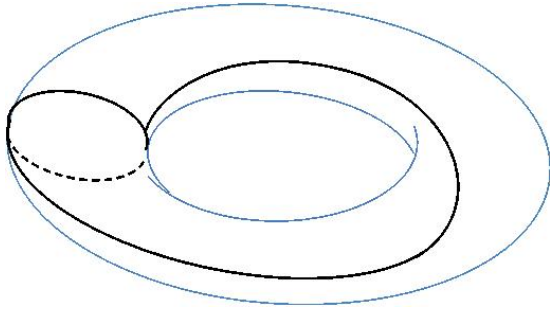
$\mathrm{GSD}^{\mathrm{disk}} = \mathrm{GSD}^{\mathrm{sphere}} = 1$ ⇨ non-chiral topological order



3 edges, 2 vertices
3 plaquetts

- Torus
  - On simplest graph:

    GSD consists of six tensors



3 edges, 2 vertices
1 plaquett

$$\mathrm{GSD} = D^{-1} \sum_{sj_1j_2j_3j_1'j_2'j_3'} d_s d_{j_1} d_{j_2} d_{j_3} d_{j_1'} d_{j_2'} d_{j_3'}$$

$$\left(G^{j_1 j_2 j_3^*}_{s j_3'^* j_2'} G^{j_3'^* j_1 j_2'}_{s j_2 j_1'} G^{j_2 j_3'^* j_1'}_{s j_1 j_3^*} \right) \left(G^{j_2^* j_3 j_1^*}_{s j_1'^* j_3'} G^{j_3' j_1'^* j_2^*}_{s j_2'^* j_1^*} G^{j_1^* j_2'^* j_3'}_{s j_3 j_2^*} \right)$$

Detect the topology of the surface by 6*j* symbols

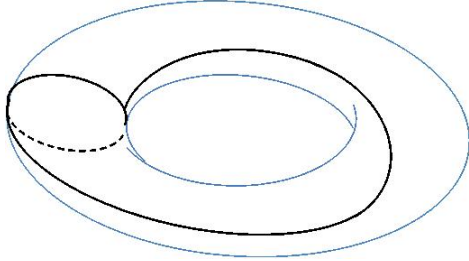
- Finite group G data on a torus:
  - Z_N group: GSD=N^2
  - General group G:

$$\mathrm{GSD} = D^{-1} \sum_{sj_1j_2j_3j_1'j_2'j_3'} d_s d_{j_1} d_{j_2} d_{j_3} d_{j_1'} d_{j_2'} d_{j_3'}$$

$$\left(G^{j_1 j_2 j_3^*}_{s j_3'^* j_2'} G^{j_3'^* j_1 j_2'}_{s j_2 j_1'} G^{j_2 j_3'^* j_1'}_{s j_1 j_3^*} \right) \left(G^{j_2^* j_3 j_1^*}_{s j_1'^* j_3'} G^{j_3' j_1'^* j_2^*}_{s j_2'^* j_1^*} G^{j_1^* j_2'^* j_3'}_{s j_3 j_2^*} \right)$$

$$\mathrm{GSD} = \left| \frac{\mathrm{Hom}(\pi_1(\mathcal{M}), G)}{G} \right|$$

Examples: on a Torus



- Quantum group $SU_k(2)$: GSD = $(k+1)^2$

  (level k: an integer) (verified up to $k = 3$)

$SU_k(2)$ has $k+1$ irreducible representations.

Compare:

SU(2) Chern-Simons theory (level k): GSD = $k + 1$

X.G. Wen, A. Zee, Phys.Rev. B58 (1998) 15717

Agrees with the conjecture that:

LW model with quantum group data is realization of doubled Chern-Simons theory on a lattice.

E. Witten, Nucl. Phys. B311, 46 (1988)

N. Reshetikhin, V. G. Turaev, Invent. Math. 103, 547 (1991)

F J Burnell and Steven H Simon, *New J. Phys.* **13**, 065001 (2011)

Summary

- Now there are discrete models on a 2d lattice (or a graph) as the Hamiltonian approach to a three-dimensional TQFT.
- There are two concrete formulations,

  which we believe are equivalent (dual) to each other
  - Generalization of Kitaev's Toric Code: gauge-theory-like
  - Levin-Wen Model (and Generalizations): purely algebraic,

    with data coming from Modular Tensor Category
- The topological properties can be worked out explicitly:
  - Ground state degeneracy on a Riemann surface
  - Fixed point wave functions of topological RG flow
  - Excitations have fractional quantum statistics etc